

Multi-Agent Platform for HNWI News Monitoring and Actionable Insights

CS480 Project Experience

PRODUCT DEMO

Overview

CONTEXT

Private banking advisors' responsibilities:

1. Conducting background research on potential private banking clients
2. Ensuring current clients are bank-compliant/in good public standing

Accomplished via accessing publicly accessible information on high net worth individuals/entities (News sources, financial blogs, auditor reports)

PROBLEM

Private banking advisors have to process lots of online information when monitoring private banking clients and conducting background research on potential clients, having to internalise news summaries and form value judgements on their own, leading to inefficiencies or cognitive biases

PRODUCT VISION

Client News Screener: efficient, reliable assistant

Multi-agentic, human-in-the-loop conversational AI that can retrieve online information on high net worth individuals, summarize online sources, and advise private bank advisors on the individual's suitability as a private banking client or to assess current client compliance

Competitor Analysis



Aggregates real-time insights from public data, uses NLP filters to put on dashboards, alerting stakeholders

Features

- Competitive intelligence with customizable alerts
- User-friendly dashboards and newsletters
- Slack, Teams, PowerBI integration



Predictive intelligence on private companies using rich data signals. Uses purpose-built AI agents to surface relevant insights

Features

- Insights on private tech companies: funding, patents, Mosaic Score predictive analytics
- VC/private equity use case and tech-focused trend detection
- Provides API access for integration



AI market intelligence search platform tapping into premium content, turns fragmented data into insight

Features

- Extensive financial content (SEC filings, broker research, expert call transcripts, trade journals, etc)
- AI search, sentiment analysis, generative AI: Smart Summaries, Deep Research
- High-quality UX, integration, support, highly rated by enterprises

Competitor Analysis - How we differ



- Inconsistent data
- Unintuitive interface
- Missing premium features



- Sole emphasis on private companies
- No generative AI features, premium features
- Limited dashboard customization



- Steep learning curve to use
- Possibly overkill for small scale client usage as opposed to financial corps

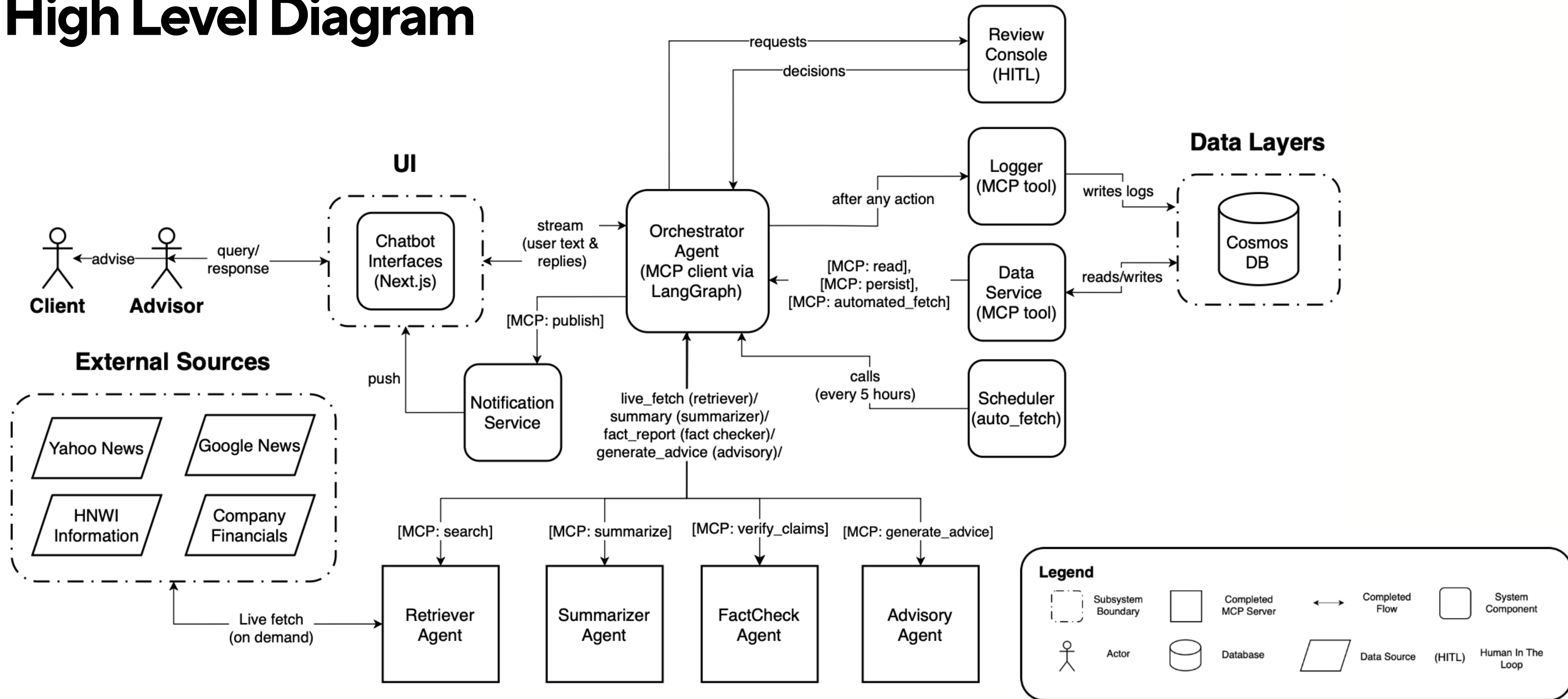
Our Selling Points

- Simple, easy to use, multi-agentic chatbot with human-in-the-loop feature
- Emphasis on publicly accessible data (including public sentiment) on high net worth individuals
- Tailored for researching on high net worth individuals' financial involvements
- Automated news retrieval alert system to provide up-to-date information on high net worth individuals

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Architecture

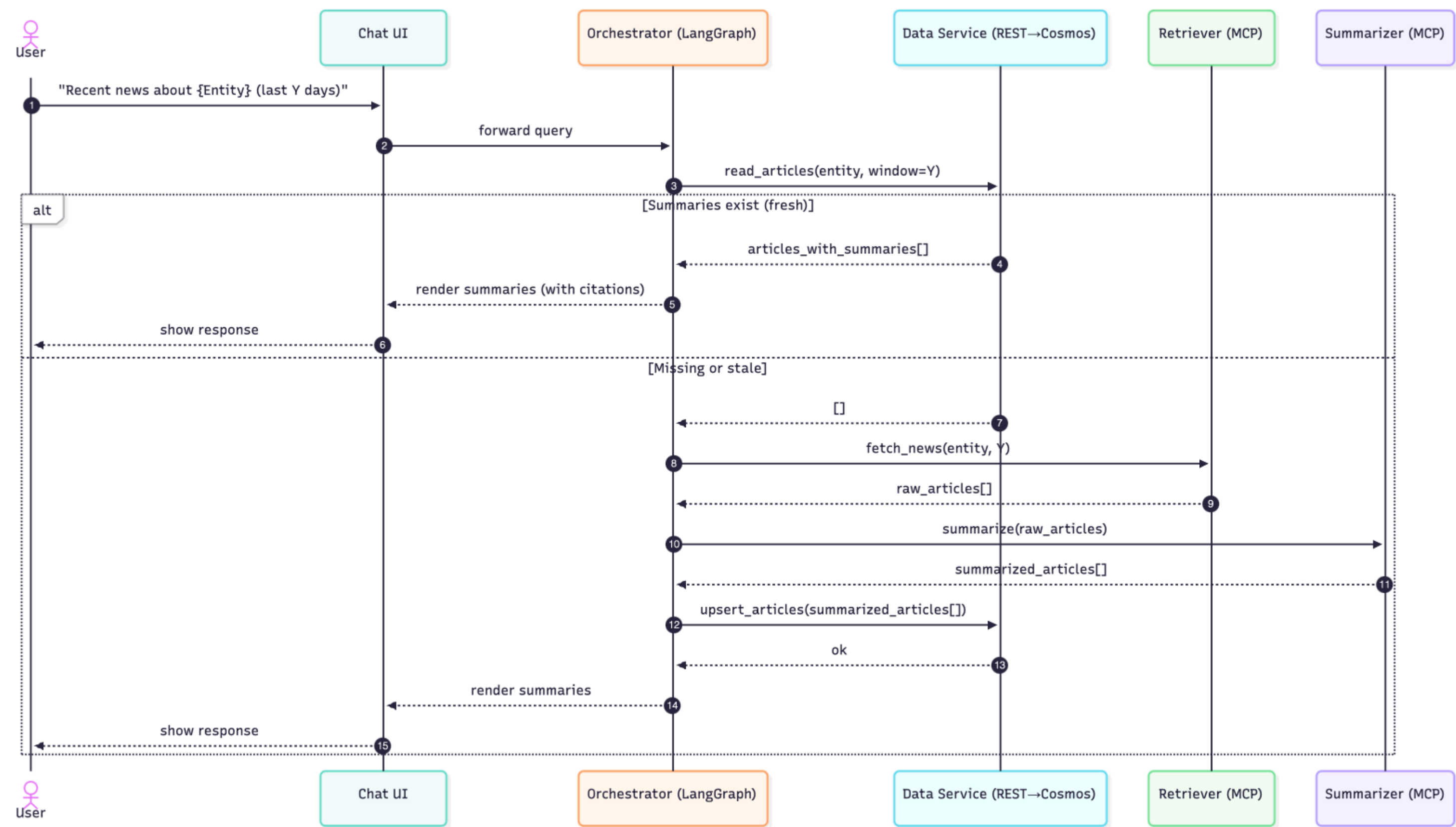
High Level Diagram



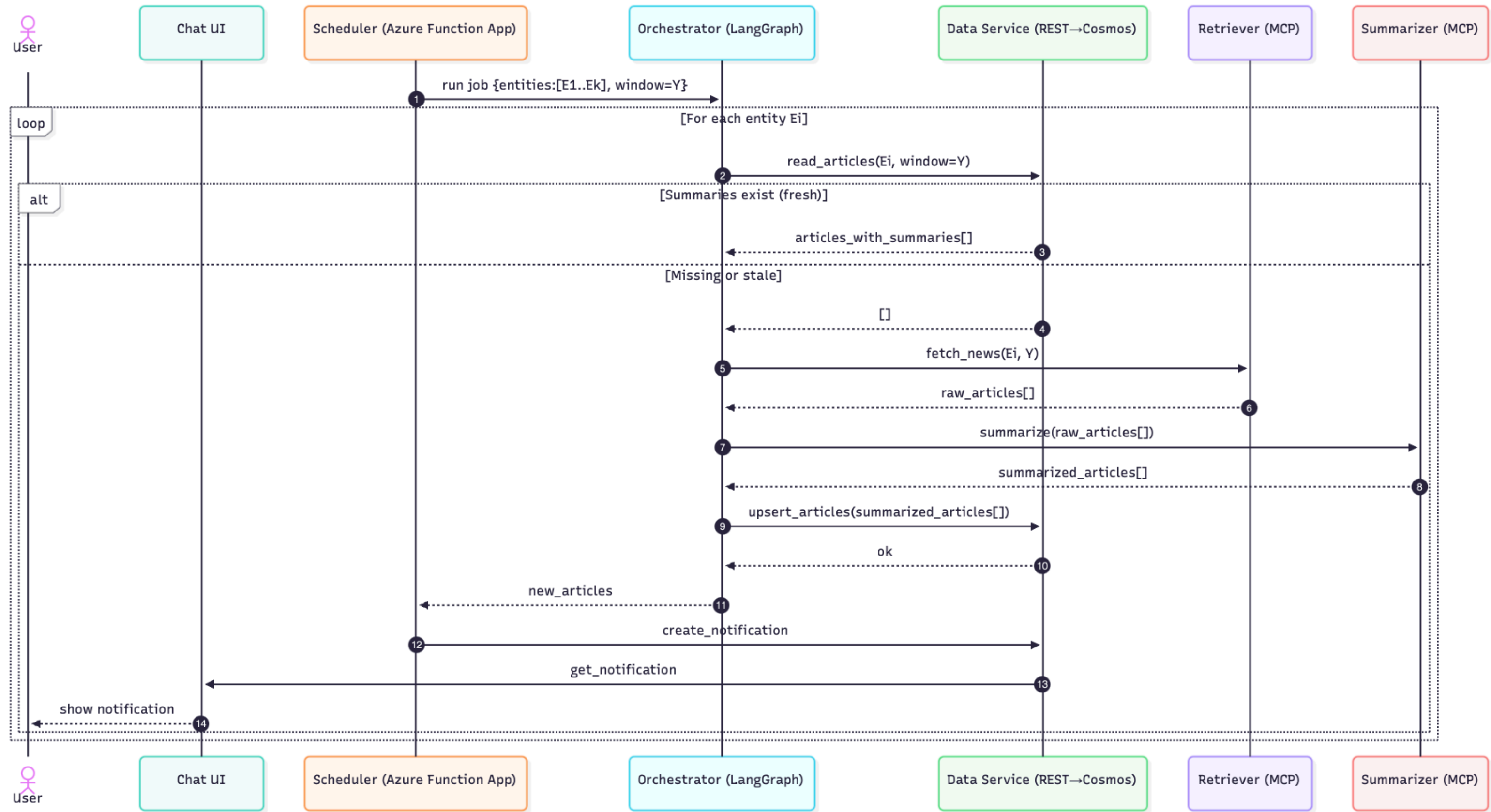
ARCHITECTURE

Sequence Diagrams

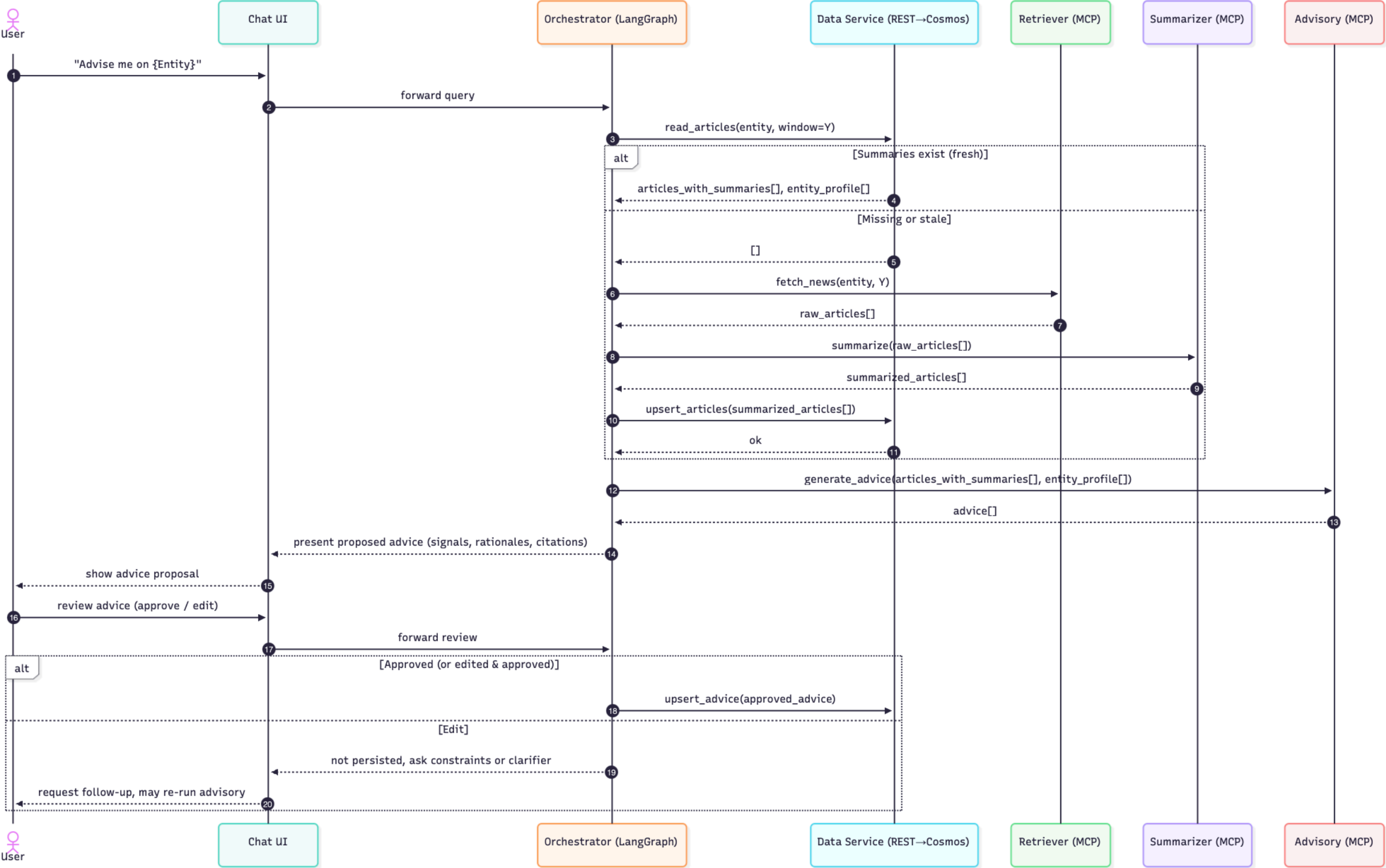
Architecture



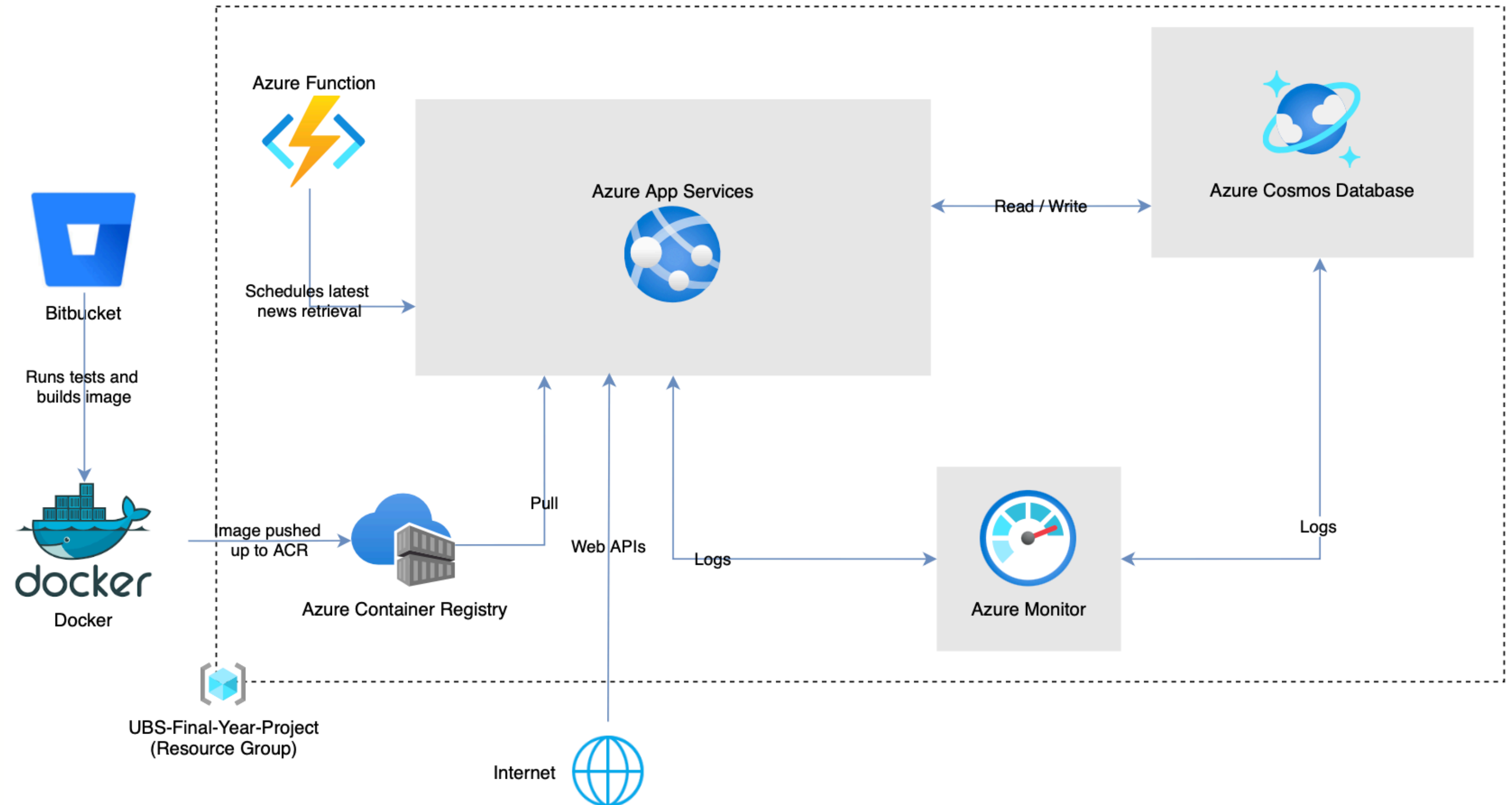
Architecture



Architecture



Infrastructure Diagram



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Implementation

Tech Stack and Deployment

Frontend

- TypeScript and Next.js
- Tailwind CSS and ShadCN UI
- Pre-commit hooks
- Automated build and deployment to Azure App Service

Backend

- Python and FastAPI
- MCP router logic, agent registry, and API endpoints
- Integrated OpenRouter and OpenAI Embedding models
- Pre-commit hooks for code quality and automated PR lint
- Built Docker image pushed to Azure Container Registry, then deployed to Azure Container Apps

Model Context Protocol (MCP)

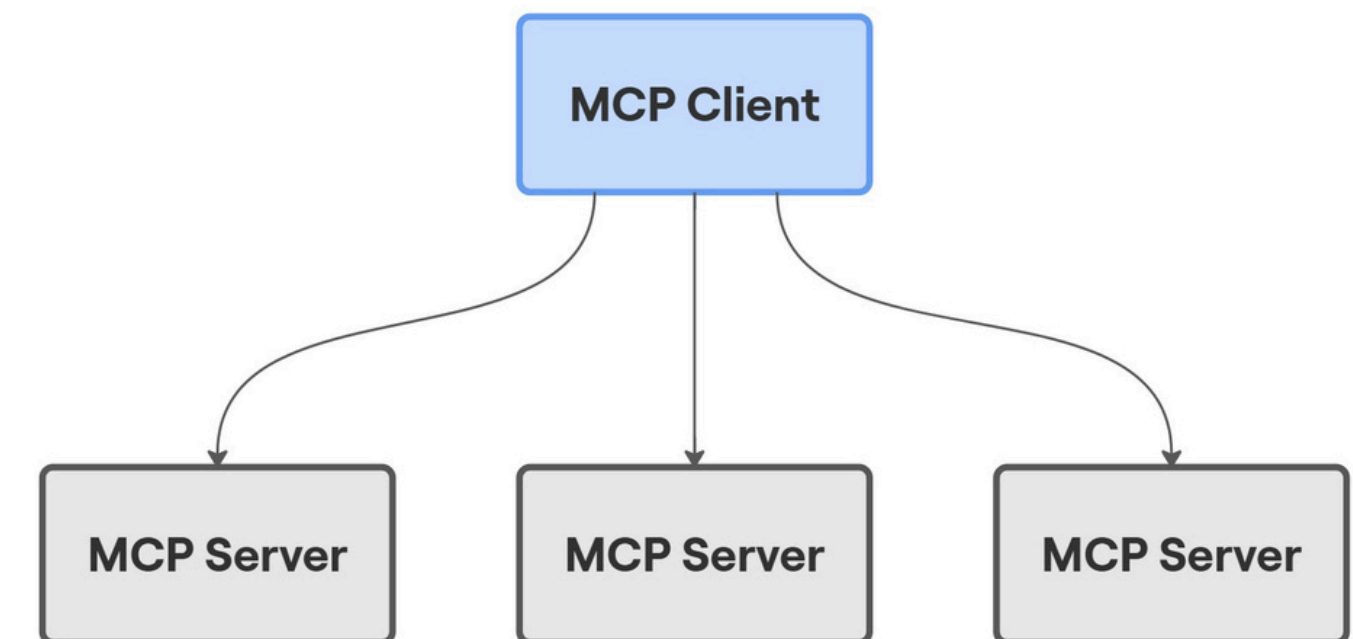
Definition

Standardized communication protocol that allows LLM-based systems to:

- Communicate with external tools in a structured way
- Connect multiple agents using a single shared JSON-RPC interface
- Ensure consistent input/output formatting

Why MCP?

- Agents return predictable JSON structures
- Multi-Agent Interoperability
- Decoupled Architecture



Model Context Protocol (MCP) - How it Works

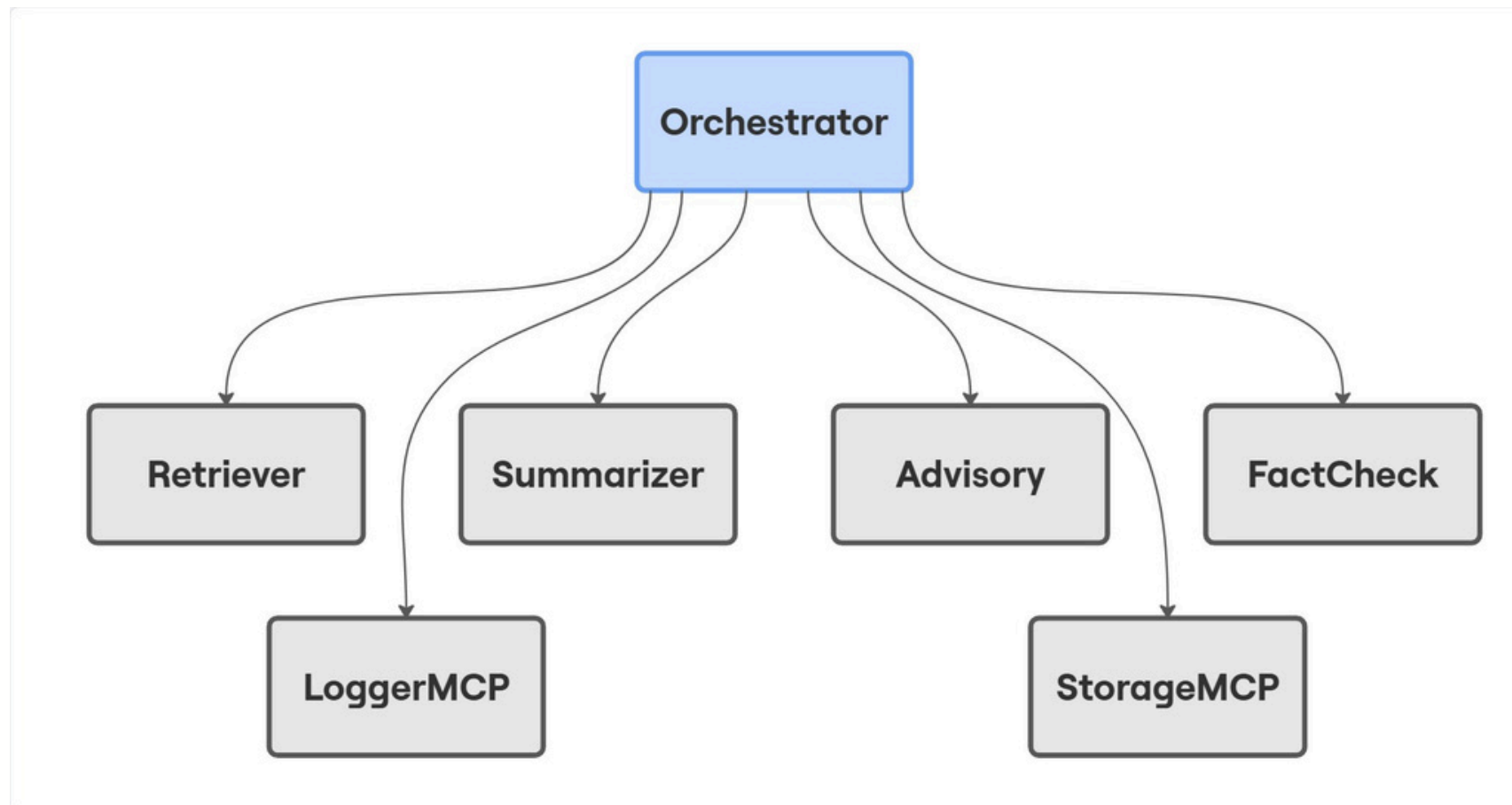
MCP Client

Orchestrator agent acts as the MCP Client: sends **JSON-RPC** requests to the MCP servers

MCP Servers

External Agents:

- Retriever Agent
- Summarizer Agent
- Advisory Agent
- FactCheck Agent
- LoggerMCP
- StorageMCP



Model Context Protocol (MCP) – Client

MCP allows Orchestrator to:

- Identify the correct agent for each intent
- Route data across agents in sequence
- Store query results in Cosmos DB through the StorageMCP tool
- Log everything systematically via LoggerMCP
- Support session memory because agent state is well-structured

Orchestrator maps servers and tool methods

Model Context Protocol (MCP) - LangGraph

MCP = communication protocol

LangGraph = workflow engine

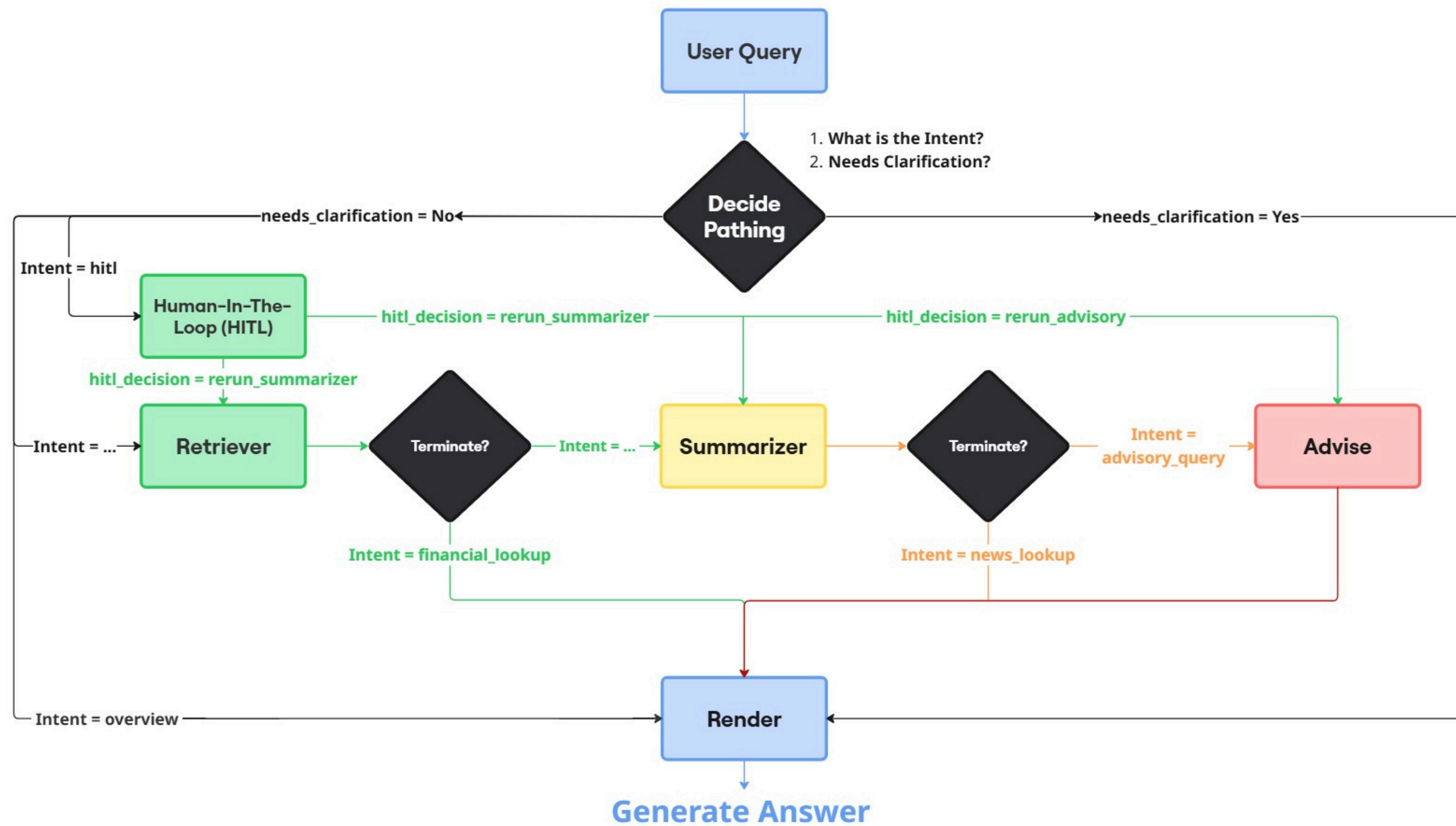
LangGraph

- Defines *when* and *in which order* they communicate (graph nodes)
- Connect nodes using edges and conditional edges

Why LangGraph?

- Enables deterministic, reproducible multi-agent pipelines
- Supports stateful sessions
- Naturally supports interruptions (e.g. HITL review)
- Makes the orchestrator easy to visualize as a state machine

Implementation



Model Context Protocol (MCP) - Servers

Each agent is an independent MCP server:

- Exposes methods using JSON-RPC
- Accepts and returns structured JSON output
- Communicates with the orchestrator via HTTP-based MCP endpoints

Retriever Agent

- Fetches news, Wikipedia/Forbes profiles, and financial data (EBITDA via Alpha Vantage)
- Purely retrieval-based, no LLM
- Fallback: Checks the DB, and if no article is found, it triggers a fresh scrape.

Summarizer Agent

- Uses an LLM to summarise articles and extract key points, including sentiment, business area, and key entities.
- Fallback: Check DB for existing summary. if found, skip summarization.

Model Context Protocol (MCP) – Servers

Advisory Agent

- Takes summaries and HNW profile; translates into actionable insights for decision-making
- Insights on controversies, investment activeness, and financial decisions

Fact Check Agent

- Generates embeddings for article summaries and finds related articles via cosine similarity.
- Compares each article with similar ones based on a similarity score to assess credibility.
- Produces a structured fact-check with status, confidence, supporting evidence, and insights using an LLM.

Model Context Protocol (MCP) – Servers

MCP Storage

- Acts as a unified data layer for all agents
- Handles structured data storage, deduplication, and retrieval via Azure Cosmos DB

MCP Logger

- Enables monitoring, debugging, and auditability of the MCP workflow
- Records events, latency, user inputs, and errors in JSON format
- Persists logs to Cosmos DB while also printing to console

Scheduled/Automated News Retriever

Workflow

1. Azure function Trigger: every x hours
2. Fetch HNWI list from Cosmos DB
3. For each HNWI, query "Recent news on <HNWI>"
4. If new articles are detected, call the /notify endpoint to save a notification in the cosmos DB

```
notification_doc = {  
    "id": notification_id,  
    "person": person,  
    "message": message,  
    "timestamp": timestamp,  
    "readers": []  
}
```

Notification Service

Frontend

- Frontend periodically polls the backend
- Backend returns only unread Notifications
- Frontend displays notification Toasts
- Frontend marks notifications as read
 - Using POST
/notifications/mark_read with its clientid

Backend

- /notify called whenever there are new news articles scraped
 - Manual query
 - Scheduled automated query

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Evaluation

Intent Routing

What we evaluated

- Built a 40-prompt test set tailored to our advisor use cases; 4-classes, 10 each
- Prompts span **easy** / **medium** / **hard** difficulty to stress edge cases

financial_lookup

public-company financial metrics

news_lookup

recent news about entity

advisory_query

client-suitability and
opportunities

smalltalk

irrelevant chat settings, non-task

How we evaluated

- Ran the orchestrator in route-only mode
- For each prompt, we captured the predicted intent and compared to self-labeled prompts
- Each session is independent of the other session

Intent Routing

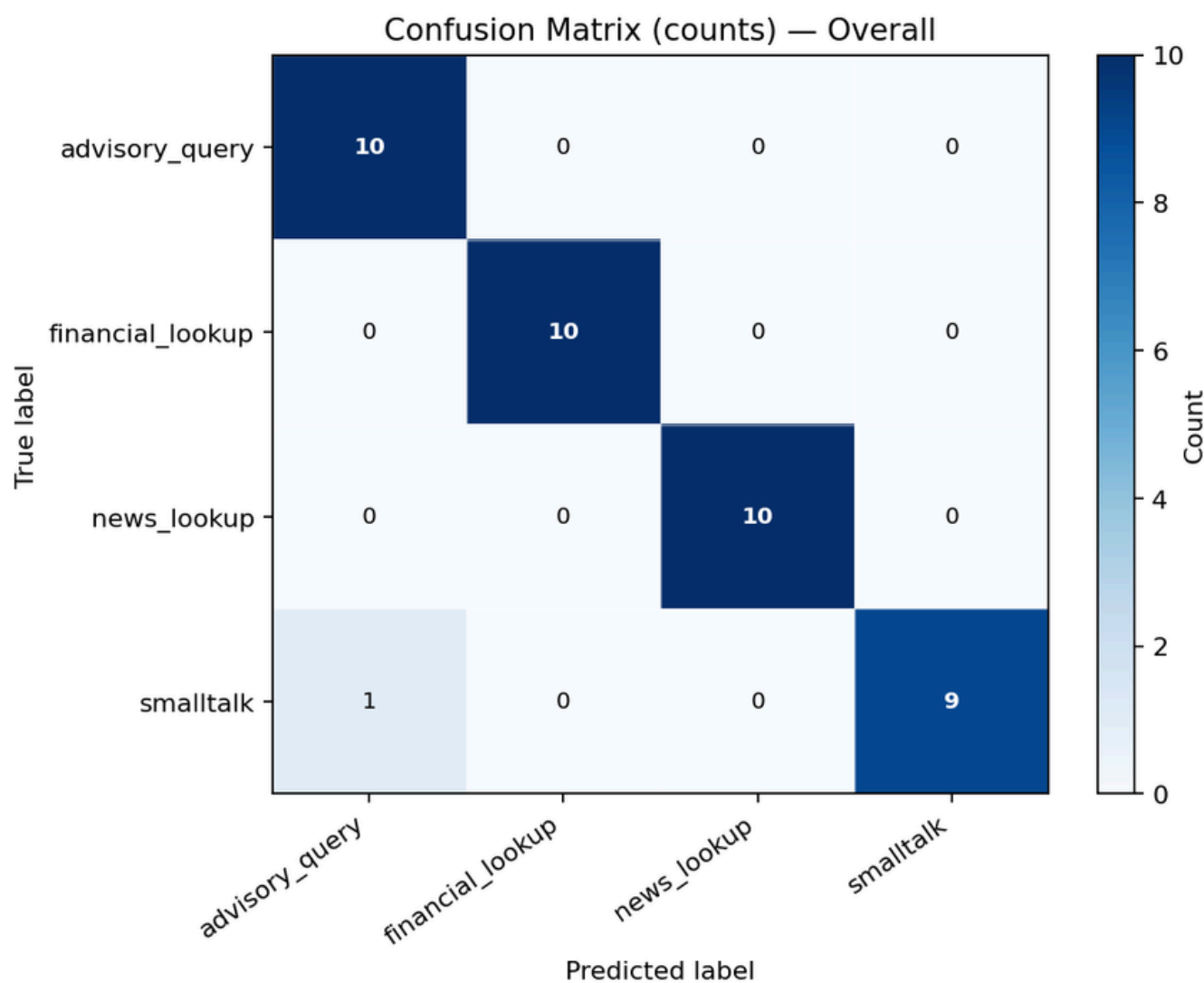
prompt	expected_intent	difficulty
financials for AAPL	financial_lookup	easy
reinvestment ideas for elon musk	advisory_query	medium
catch me up on Microsoft layoffs rumors this week	news_lookup	hard

Sample Test

pred_intent
financial_lookup
smalltalk
news_lookup

Sample Prediction

Intent Routing - Overall Results



Confusion Matrix

	precision	recall	f1-score	support
advisory_query	0.909	1.000	0.952	10
financial_lookup	1.000	1.000	1.000	10
news_lookup	1.000	1.000	1.000	10
smalltalk	1.000	0.900	0.947	10

accuracy			0.975	40
macro avg	0.977	0.975	0.975	40
weighted avg	0.975	0.975	0.975	40

Classification Report

Summarization Quality

Context

- Articles are freshly scraped → no human reference for summaries
- We evaluate reference-free metrics that align with advisor needs

Factuality

Consistency of summary statements with the source

Semantic Coverage

Meaning overlap with the article

Redundancy

Non-repetitiveness of the summary text

Readability

Reading grade level

Brevity

Absolute tokens/chars in the summary

Compression

compactness

Summarization Quality

Context

- Articles are freshly scraped → no human reference for summaries
- We evaluate reference-free metrics that align with advisor needs

FactCC

Consistency of summary statements with the source

BERTScore

Meaning overlap with the article

n-gram Ratio

Non-repetitiveness of the summary text

Flesch-Kincaid Grade

Reading grade level

Summary Length

Absolute tokens/chars in the summary

Compression Ratio

compactness

Summarization Quality - Results

model	factcc_score	bertscore	2-gram_ratio	fk_grade	summary_length	compression_ratio
deepseek-chat-v3.1	0.754	0.895	1.000	13.836	74.667	0.658
gemini-2.5-flash	0.044	0.863	0.996	12.485	69.000	0.381
llama-4-maverick	0.066	0.852	1.000	11.847	58.333	0.338
mai-ds-r1	0.346	0.852	1.000	12.936	65.667	0.374
qwen3-4b	0.307	0.850	1.000	13.432	61.000	0.350

deepseek best matched our priorities

- high factual consistency
- strong semantic alignment

Advisory Generation - DeepEval

DeepEval

Open-source LLM evaluation framework; For our advisory output:

- Runs advisory test cases through predefined metrics
- Uses LLM-as-a-Judge (DeepSeek Terminus via OpenRouter) to score each output
- Computes quantitative scores (0-1) on correctness, completeness, and answer relevance

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Correctness

Check if the reasoning, claims, and conclusions match the ground-truth “golden” answer

Completeness

The judge model compares the actual answer to the expected one and checks if important elements are missing

Answer Relevancy

Detects irrelevant statements, off-topic reasoning, hallucinated or unnecessary details

Stores the results on Confident AI

Advisory Generation - Confident AI

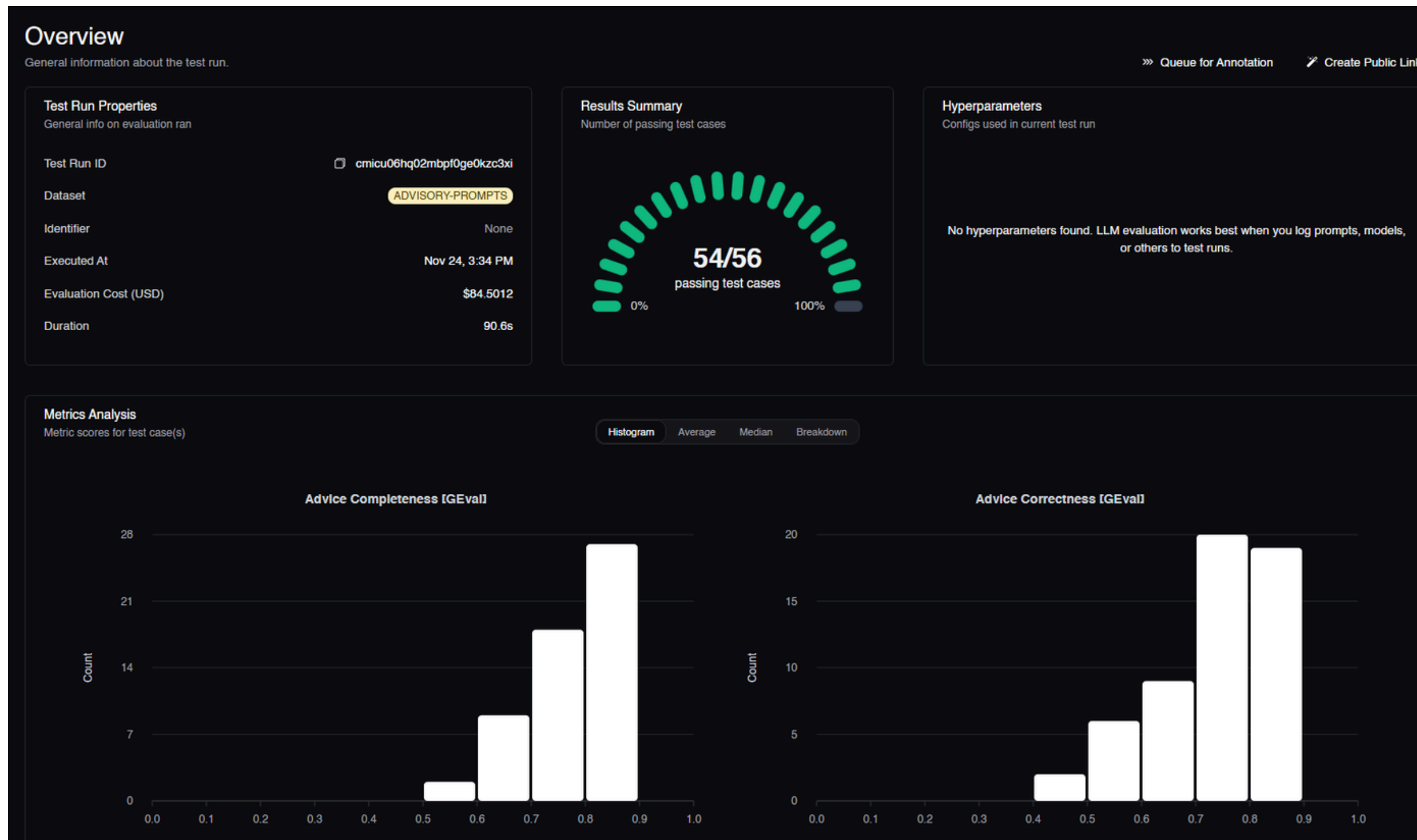
Confident AI

Platform for storing datasets, running evaluations, and creating 'goldens'

- Store advisory evaluation dataset
- Uses DeepSeek Terminus to auto-generate "expected" advisory output
- Run manually, but can be integrated with pytest

```
metrics:
  advisory_completeness:
    type: geval
    name: Advice Completeness
    criteria: |
      Assess the completeness of the information provided in the actual output based on the information available in the input prompt.
      Award a higher score for actual outputs that have managed to correctly extract most information out from the input prompt,
      and assign lower scores for outputs that neglect information that could be extracted from the input prompt.
    evaluation_params:
      - INPUT
      - EXPECTED_OUTPUT
      - ACTUAL_OUTPUT
    threshold: 0.5
    model: openrouter_deepseek
```

Advisory Generation - Confident AI Overview



Advisory Generation - Confident AI Details

Test Case Details

- 3 main metrics used
 - Advice Completeness
 - Advice Correctness
 - Answer Relevancy
- Scores assigned from 0 to 1
- Used stronger model (GPT-4o-mini) for comparing expected and actual advisory output
- Provides qualitative feedback on actual output to improve advisory system prompt

Test Case Details		✕
Test Case ID	d86f323f-e965-4bba-be50-deb3b13612aa	
Status	Success	
Run Duration	24.5s	
Metrics	✓ Advice Completeness [GEval] ✓ Advice Correctness [GEval] ✓ Answer Relevancy	
Metrics Data	Test Case Parameters	Additional Info
Metric Name	Advice Completeness [GEval]	
Status	Success	
Evaluation Model	openrouter/openai/gpt-4o-mini (('openai/gpt-4o-mini', 'openrouter', None, None))	
Score	0.88	Threshold 0.5
Reason	The actual output closely aligns with the expected output, accurately reflecting the key information from the input prompt. It correctly identifies that no information is available regarding Lisa Su's net worth, investment portfolio, or investment activity, which is consistent with the articles reviewed. The associations section is comprehensive, listing relevant companies and individuals with appropriate citations. The risk assessment accurately notes the absence of controversies and assigns low risk ratings based on the positive portrayal of Lisa Su in the articles. The suitability analysis provides a strong justification for the overall rating and service usage likelihood, though the likelihood rating is slightly lower than expected, indicating a minor deviation from the expected output.	

Pytests

```
----- coverage: platform win32, python 3.12.11-final-0 -----
Name                               Stmts  Miss  Cover   Missing
-----
apps\common\enums.py                38      0   100%
apps\common\memory_session.py       70      0   100%
apps\common\paths.py                44      6    86%   51, 56, 100-104
apps\common\yaml_config.py          14      0   100%
-----
TOTAL                               166      6    96%
Coverage HTML written to dir htmlcov

Test tests/paths_test.py::TestPathResolutionEdgeCases::test_paths_work_with_symlinks was skipped. Reason:
('C:\\Users\\samue\\Desktop\\fyp-ubs-backend\\tests\\paths_test.py', 259, 'Skipped: Cannot create symlinks on this
system')
```

Pytest to evaluate common utilities used in the codebase

- On average 96% test coverage for all the functions implemented
- Coverage report generated after every pytest run

Performance Runtime

Retriever, Summarizer, and Fact Checker

- Typical runtime: 60 - 120 seconds
- Duplicate articles found in DB: 10-20 seconds

Advisory

- Typical runtime: 120 - 180 seconds depending on number of sources considered

Combined Runtime

- ~5 minutes

Acceptable, within sponsor's expectations

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DEMO

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Project Management

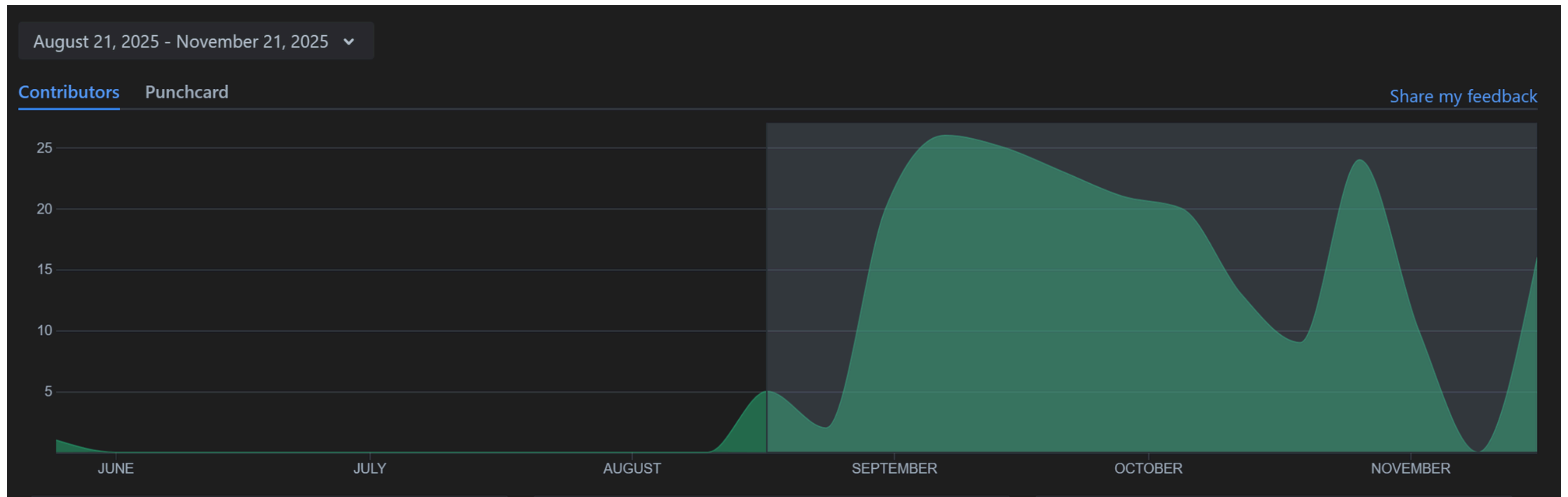
Jira

☐	Work
☐	⚡ FYP-110 Notification Service
☐	⚡ FYP-94 Data Service
☐	⚡ FYP-77 Fact Check Agent
☐	⚡ FYP-73 Chatbot Frontend
☐	⚡ FYP-7 Production and Demos
☐	⚡ FYP-5 Orchestrator Agent
☐	⚡ FYP-4 Retriever Agent
☐	⚡ FYP-3 Advisory Agent
☐	⚡ FYP-2 Summarizer Agent
☐	⚡ FYP-1 Setup & Architecture

- We organized our development work using Epics, each containing multiple user stories that map directly to system functionality and stakeholder requirements
- Each major component was structured as its own Epic, giving us a total of 10 Epics for the entire project

PM

Source Control (Bitbucket)

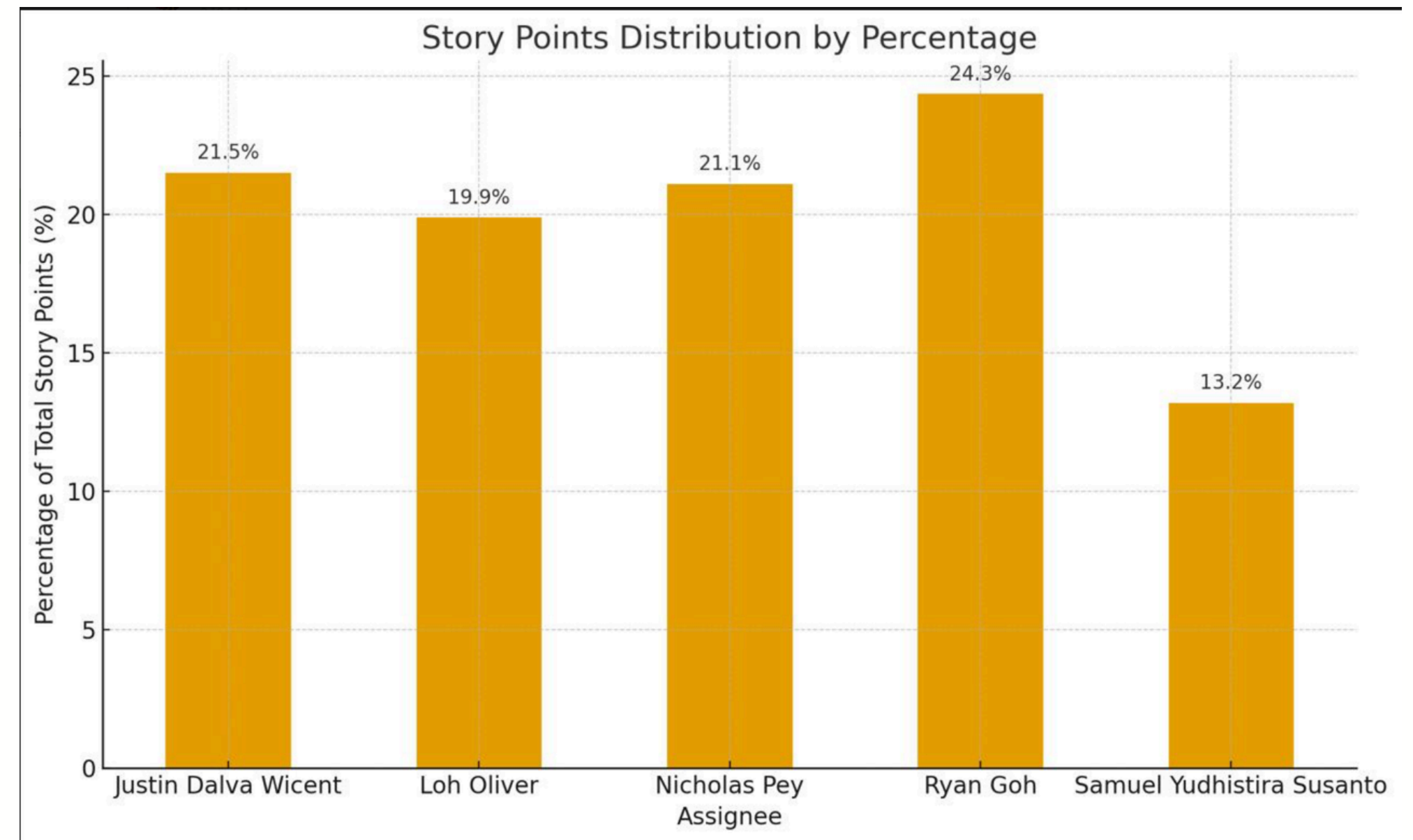
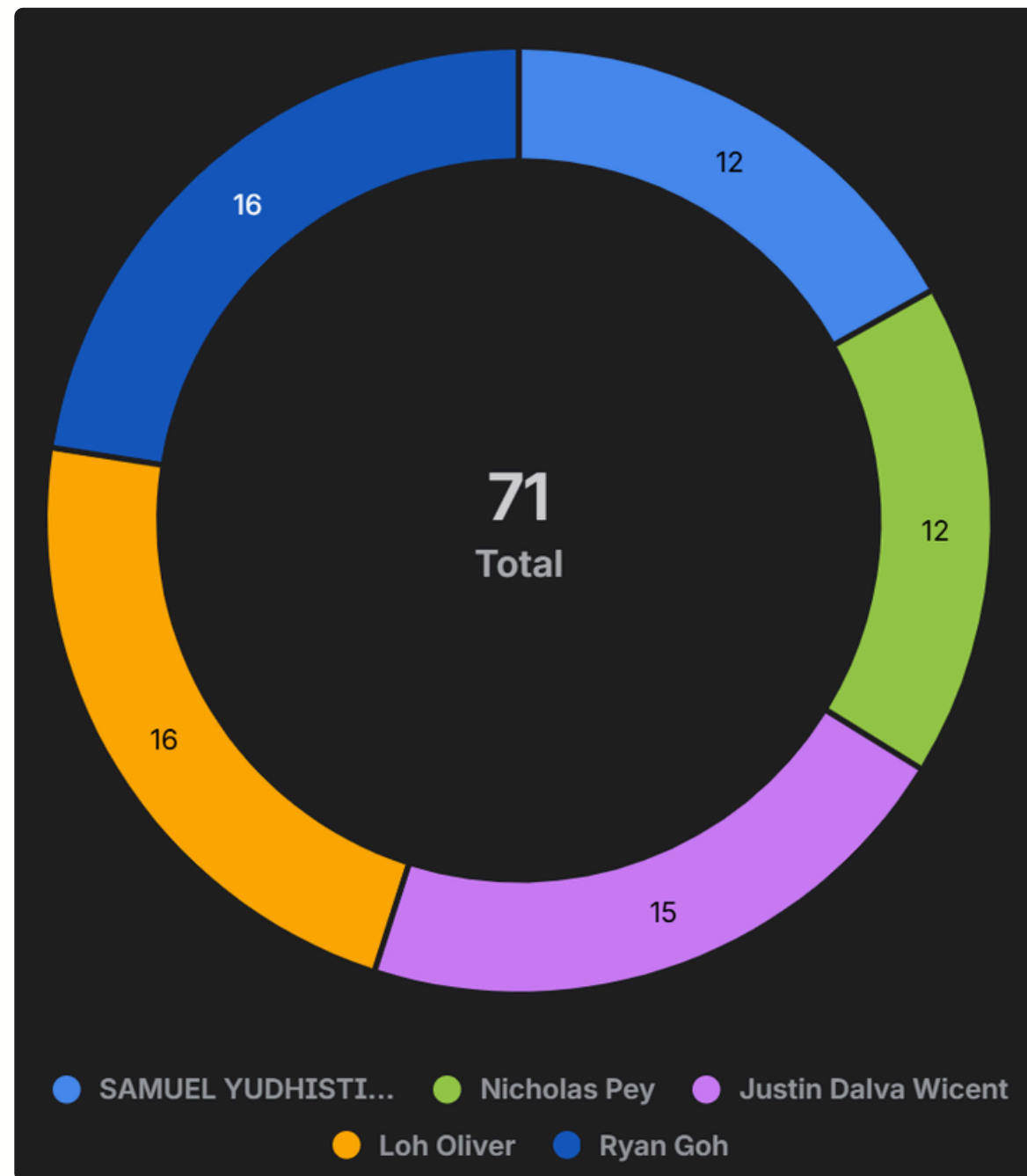


Bitbucket's commit history

Source Control (Bitbucket)

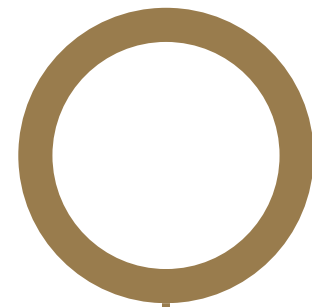


Work Distribution



Members' work distribution by story points

Sprint Progress

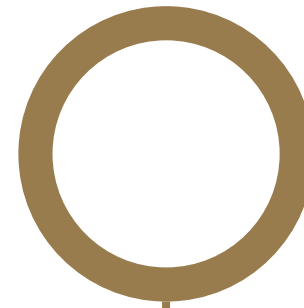


Sprint 6

Notification feature
intialization

Agent and common
utilisties testing

Front-end page
development

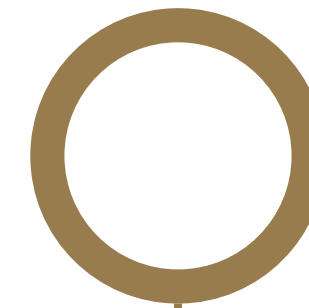


Sprint 7

Notification feature
finalization

Agentic AI testing

HITL and workflow
refinement



Sprint 8

Final deployment of
backend and frontend
services

Session, chat, source,
and flow refinement

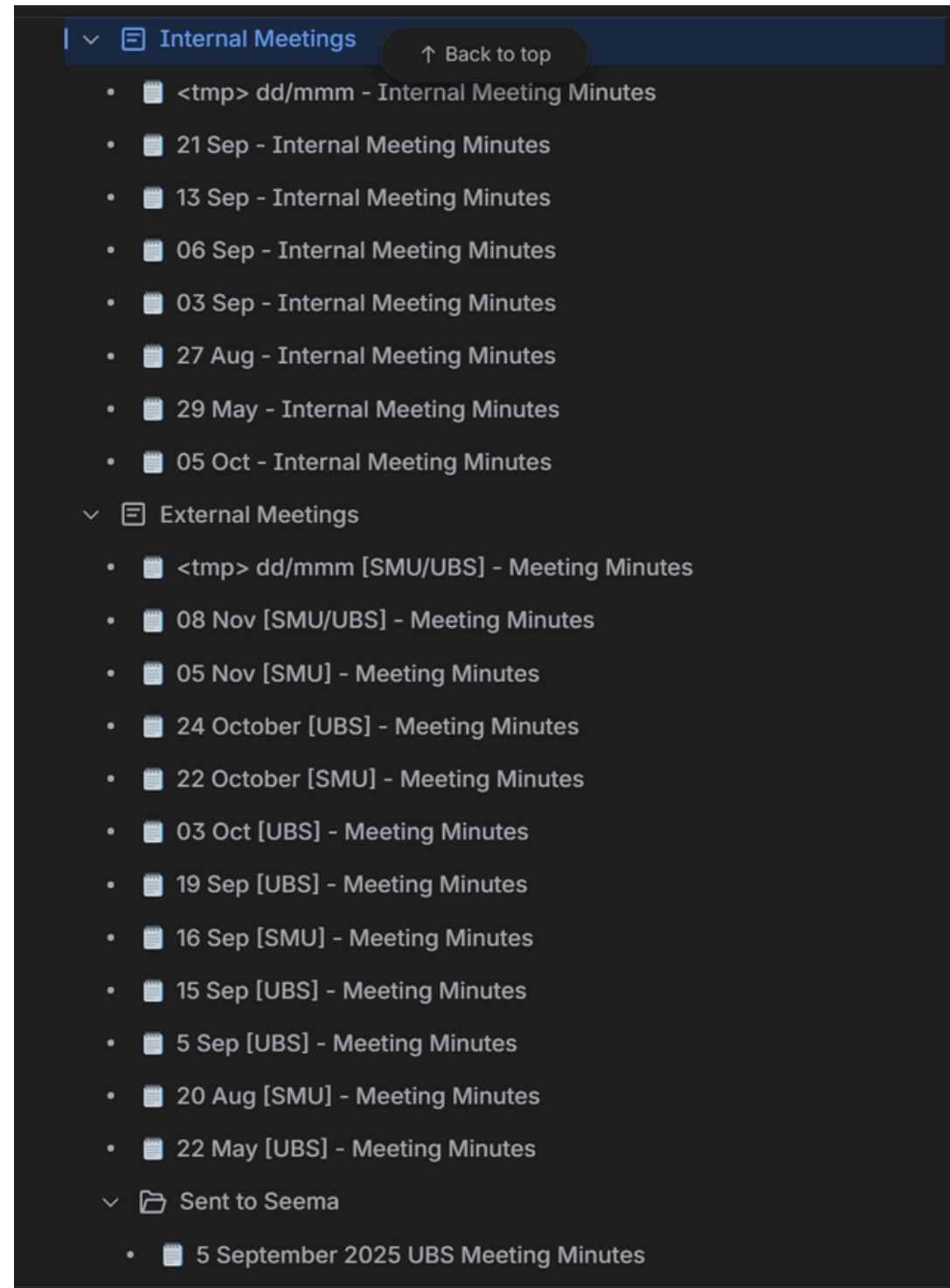
Final deliverables
preparation

Documentation - Confluence Structure

- - ├─ Proposal Documents
 - ├─ Meeting Minutes/
 - ├─ Internal Meetings
 - └─ External Meetings
 - ├─ Research/Knowledge Sharing
 - ├─ Repository Developments
 - ├─ General
 - └─ Miscellaneous

- Every meeting is recorded in a meeting minute, written and documented inside Confluence
- Every new implemented feature, graphs, and research report is also documented and sorted

Documentation - Meeting Minutes



- Regular internal and external meetings conducted
- Minutes documented on Confluence for every session
- Aligned progress with client expectations
- Facilitated decision-making and task tracking
- Gathered more insights and feedback both from our supervisors and UBS (Seema)

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Conclusion

Current and Future Works

What we've achieved

- Conversational AI with full core functionalities
- Robust Orchestrator With Dynamic LangGraph Workflow
- Fully stateful sessions with memory and context reuse
- Modular MCP servers with clean JSON-RPC interfaces (industry-standard protocol)
- Reproducible LLM-as-Judge scoring pipeline for evaluation
- Clean and intuitive UI
- Full deployment with CI

Current and Future Works

Our Long-Term Vision for This Platform

- Adopt locally fine-tuned LLMs
- Transition fully into a microservice architecture
- Move to more scalable Agentic AI frameworks
- Strengthen CI/CD and observability
- Enhance system modularity

There are lots of room to grow!

Thank you!